From Conditional Monte Carlo Diagnostics to FTV-Calibrated Consistent Rollout Models

Progress update

Abstract

I used the conditional Monte Carlo experiment to stress-test the previously successful consistent graph rollout model under the way I ultimately want to use it: patient-level uncertainty estimation and treatment-response reasoning. The original scheduled-sampling model produced coherent graph rollouts, but the Monte Carlo analysis revealed a specific biological calibration problem. The deterministic T0-to-T3 rollout underpredicted mean functional tumor volume (FTV) by 12.51 mL, and the residual Monte Carlo correction overcompensated by moving the Monte Carlo mean above the observed mean. Raw 90% intervals under-covered, and conformal intervals reached nominal coverage only after substantial widening. These findings motivated a targeted retraining grid with aggregate FTV and alive-burden objectives. The best retrained model reduced T0-to-T3 FTV MAE from 12.88 mL to 5.88 mL while leaving spatial metrics essentially unchanged. I then reran the same conditional Monte Carlo sampler on the retrained checkpoints. For the primary retrained model, T0-to-T3 MC-mean FTV MAE dropped from 15.26 mL to 7.61 mL, raw 90% coverage increased from 0.766 to 0.876, conformal 90% interval width shrank from 59.51 mL to 27.95 mL, and MC alive-count error dropped from 40.78 to 2.69 supervoxels. The interpretation is that the Monte Carlo experiment did not invalidate the earlier rollout model; it identified the next modeling requirement: endpoint-calibrated tumor burden before uncertainty estimates can be trusted for response analysis. The conclusion from this phase is that `bio_ftv020_alive005` is the best balanced candidate to carry forward. Its predictions can now support patient-level imaging questions that the original rollout model could not answer reliably: what future MRI FTV is likely after conditioning on available visits, how wide the plausible FTV range is, whether the observed endpoint falls inside a calibrated interval, and how much active tumor burden remains in the graph. That is meaningful because the model is no longer only a plausible spatial forecaster; it is becoming an endpoint-calibrated uncertainty model for MRI tumor-burden response analysis.

1 Why this experiment mattered

The previously retained model, `v2_sched_samp`, was already a strong consistent graph rollout model. It used scheduled sampling and learned to roll registered tumor graphs through the I-SPY2 imaging sequence:

$$T0 \to T1 \to T2 \to T3.$$

Each graph node represents a tumor supervoxel with a 3D position, imaging features, a volume-like feature, and an alive probability. The model predicts how those node states evolve across treatment.

That model was useful as a spatial rollout model, but I wanted to know whether it could also serve as the center of a patient-level uncertainty distribution. This is a harder question. A graph trajectory can look plausible while still being wrong about total active tumor burden. For treatment-response modeling, that distinction matters because FTV is the clinically interpretable endpoint that connects the graph prediction to tumor response.

2 Measurement and modeling terminology

Several measurements recur throughout this update. They are all related, but they answer different questions about the rollout model.

Functional tumor volume. Functional tumor volume (FTV) is the estimated volume of active enhancing tumor tissue on breast MRI. In this work, it is computed from the predicted or observed supervoxel feature that tracks active tumor volume. FTV is important because it is the patient-level imaging endpoint that most directly represents active tumor burden during neoadjuvant treatment. A model can produce a reasonable-looking tumor cloud while still predicting too little or too much total active tumor. That is why FTV calibration became the central issue after the Monte Carlo analysis.

Near-zero T3 MRI FTV diagnostic. The rollout model predicts an imaging trajectory through the available MRI time points. The Monte Carlo script reported

$$P(\mathrm{FTV}_{T3} < 0.1\ \mathrm{mL}),$$

which answers one imaging question: how often does the simulated T3 MRI tumor burden become essentially zero under this threshold? I treat this only as a descriptive MRI FTV diagnostic. It is not used as a training target, loss term, or primary endpoint in the retraining analysis.

Supervoxels, alive probability, and alive mass. The tumor graph is made of supervoxels, which are small 3D regions of tumor tissue. Each supervoxel has a predicted alive probability at the next visit. Alive mass is the sum of those probabilities across the graph:

$$\text{alive mass} = \sum_i P(\text{node } i \text{ alive}).$$

The observed alive count is the number of supervoxels that are actually present or active at the observed future visit. Alive mass matters because predicted FTV is not just the sum of all predicted node volumes; it should be the sum of active volume from nodes that remain alive.

FTV residuals and log-volume residuals. The conditional Monte Carlo sampler is residual based. An aggregate FTV residual is the difference between observed FTV and deterministic predicted FTV:

$$\text{FTV residual} = \text{observed FTV} - \text{predicted FTV}.$$

This residual adjusts the sampled patient-level tumor burden. Log-volume residuals are node-level residuals computed in log space:

$$\log(1 + \text{observed node volume}) - \log(1 + \text{predicted node volume}).$$

Log space is used because tumor volumes have a long right tail; it avoids letting a small number of very large nodes dominate the residual distribution.

Spatial residuals and cloud metrics. The sampler also uses spatial residuals. A global displacement residual shifts the whole predicted tumor cloud coherently, while local displacement residuals add node-level spatial variation. Spatial quality is evaluated with sliced Wasserstein distance (SWD), Chamfer distance, and Dice. SWD and Chamfer measure how far the predicted and observed point clouds are from each other; lower is better. Dice measures overlap after voxelizing the clouds; higher is better.

Uncertainty metrics. Coverage measures how often the observed FTV lands inside the model's predicted uncertainty interval. Raw 90% coverage uses the Monte Carlo FTV distribution directly: the lower endpoint is the 5th percentile and the upper endpoint is the 95th percentile. If the model is well calibrated, about 90% of held-out observations should fall inside those intervals. Conformal 90% coverage starts from the same raw Monte Carlo interval but then expands it using held-out nonconformity scores, which are the observed misses between predicted intervals and true FTV values. This gives an empirical correction when the raw Monte Carlo intervals are too narrow or systematically off center. Interval width measures the size of those uncertainty intervals, so it captures sharpness: narrower intervals are better only when coverage remains adequate. The Continuous Ranked Probability Score (CRPS) evaluates the entire predicted FTV distribution against the observed FTV, not just one interval or one point estimate. Lower CRPS means the distribution puts more probability mass near the observed value while still remaining appropriately spread out.

3 Conditional Monte Carlo experiment design

The Monte Carlo experiment was designed as a stress test of whether a deterministic graph rollout could be used as the center of a clinically useful uncertainty distribution. The deterministic model gives one predicted future tumor graph. The Monte Carlo layer asks a different question: given the errors seen in held-out patients, what range of future FTV values and tumor clouds are plausible for a new held-out patient?

Base model and held-out protocol. The original experiment used the retained scheduled-sampling consistent rollout checkpoints:

`runs/consistent_forecaster_v2/v2_sched_samp`.

Each patient was evaluated with the held-out fold checkpoint for that patient, so the checkpoint used for a patient was not trained on that patient. The full run included 758 patients, 4,548 rollout records, and 1,164,288 Monte Carlo samples, with 256 samples per rollout record.

The same accounting was used in the retrained follow-up. Each retrained model produced 4,548 rollout records and 1,164,288 MC samples. Because I ran three retrained checkpoints, the follow-up generated 3,492,864 retrained MC samples. Including the original baseline run used for comparison, the analysis compared 4,657,152 MC samples across the baseline and retrained models. The sample count comes from

$$758 \text{ patients} \times 6 \text{ conditioning buckets} \times 256 \text{ MC draws} = 1{,}164{,}288 \text{ samples per model.}$$

Conditioning design. The experiment was conditional because it treated each patient's already observed visits as known information, then sampled uncertainty only over the future visits. This matches the clinical use case: at different points during treatment, different amounts of imaging history are available. The experiment evaluated three conditioning regimes:

- T0 observed, then predict T1, T2, and T3.
- T0 and T1 observed, then predict T2 and T3.
- T0, T1, and T2 observed, then predict T3.

This produced six start-visit/predicted-visit buckets: T0→T1, T0→T2, T0→T3, T1→T2, T1→T3, and T2→T3. The T3 buckets were the most important because T3 is the final pre-surgery MRI time point. Because every patient contributes all six buckets, each model has $758 \times 6 = 4{,}548$ patient/bucket rollout records before MC sampling.

Calibration residual design. The sampler was residual based. First, the deterministic rollout was run for each patient and each conditioning bucket. Then the difference between the deterministic prediction and the observed held-out outcome was stored as a calibration residual. Residuals were bucketed by start visit and predicted visit, because the error distribution is not the same for T0→T1 as it is for T0→T3. For each target patient, that patient's own residuals were excluded from the calibration pool. This made the sampling leave-one-patient out with respect to residual information: the patient being evaluated could not contribute its own error back into its uncertainty distribution.

For a given target patient and bucket, the patient-level residual candidates are therefore the other 757 patients in that same bucket. This exclusion is important because it prevents a patient from borrowing its own observed error when constructing its MC distribution.

What one Monte Carlo draw did. Each Monte Carlo draw started from the deterministic held-out-fold rollout for one target patient and one conditioning bucket. The model weights, observed conditioning visits, fold assignment, and deterministic rollout center were not randomized. What changed across draws was the residual information resampled from other patients in the same bucket. Each draw randomized:

1. one patient-level calibration residual, sampled with replacement from the non-target patients in the same start-visit/predicted-visit bucket;
2. the aggregate FTV residual from that sampled calibration patient, defined as observed FTV minus deterministic predicted FTV;
3. the alive-count residual from that sampled calibration patient, defined as observed alive count minus deterministic alive mass;
4. the global displacement residual from that sampled calibration patient, which shifts the predicted tumor cloud coherently;
5. an alive mask, sampled by Gumbel top-k from the target patient's predicted alive probabilities after the sampled alive-count residual sets the total number of alive nodes;
6. local displacement residuals, sampled independently with replacement for each node from the pooled node-level residuals in the same bucket; and
7. log-volume residuals, sampled independently with replacement for each node from the pooled node-level volume residuals in the same bucket.

For each draw, the sampler perturbed positions by adding the sampled global shift plus node-level local displacement noise. It perturbed node volumes in log space by applying sampled log-volume residuals, then clipped sampled volumes to be nonnegative. Sampled FTV was computed as the sum of sampled volumes over the sampled alive nodes, plus the sampled aggregate FTV residual, clipped at zero. Repeating this 256 times gave an empirical distribution of possible future FTV values and tumor-cloud summaries for that patient/bucket rather than a single deterministic rollout.

Why the sampler was held fixed. The purpose of the first Monte Carlo experiment was diagnostic, not to tune the sampler until the final numbers looked better. I therefore kept the sampling procedure fixed and asked whether the uncertainty distribution exposed a weakness in the deterministic rollout center. In the retrained follow-up, I kept that same sampler fixed again and swapped only the base model. This isolated the question I cared about: if the deterministic FTV center is corrected, do the Monte Carlo mean, intervals, coverage, CRPS, and spatial sample summaries become better calibrated without changing the uncertainty machinery itself?

Outputs and evaluation. For every rollout record, the experiment saved the deterministic prediction, the Monte Carlo sample distribution, per-patient summaries, and calibration summaries. The main FTV outputs were deterministic bias and MAE, Monte Carlo mean and median bias/MAE, raw 90% coverage, conformal 90% coverage, interval width, and Continuous Ranked Probability Score (CRPS). The main graph outputs were alive-count error, sliced Wasserstein distance (SWD), Chamfer distance, and Dice. This combination was important because a model can have acceptable-looking spatial rollouts while still being biologically miscalibrated in total tumor burden, or it can improve FTV while degrading the tumor cloud geometry.

4 What the baseline Monte Carlo simulation revealed

This section refers to the original conditional MC run on the baseline `v2_sched_samp` scheduled-sampling model, before biological retraining. It is the diagnostic experiment that motivated the retraining, not the final retrained MC comparison. The first important result was systematic deterministic FTV underprediction. The T0-to-T3 endpoint was the most important case:

$$\text{observed mean T3 FTV} = 24.94 \text{ mL}, \qquad \text{deterministic mean T3 FTV} = 12.43 \text{ mL}.$$

That is a mean signed error of -12.51 mL and an FTV MAE of 12.88 mL. In biological terms, the model learned a rollout trajectory that shrank active tumor burden too aggressively.

For the baseline model, the residual Monte Carlo correction did not fix the problem as a point predictor. It corrected the direction of the bias but overcorrected the magnitude. For T0-to-T3, the Monte Carlo mean was 31.30 mL against the observed mean of 24.94 mL, giving a +6.36 mL bias and a 15.26 mL MC-mean MAE. That told me the uncertainty layer was partly compensating for a biased base model rather than expressing uncertainty around a well-centered deterministic prediction.

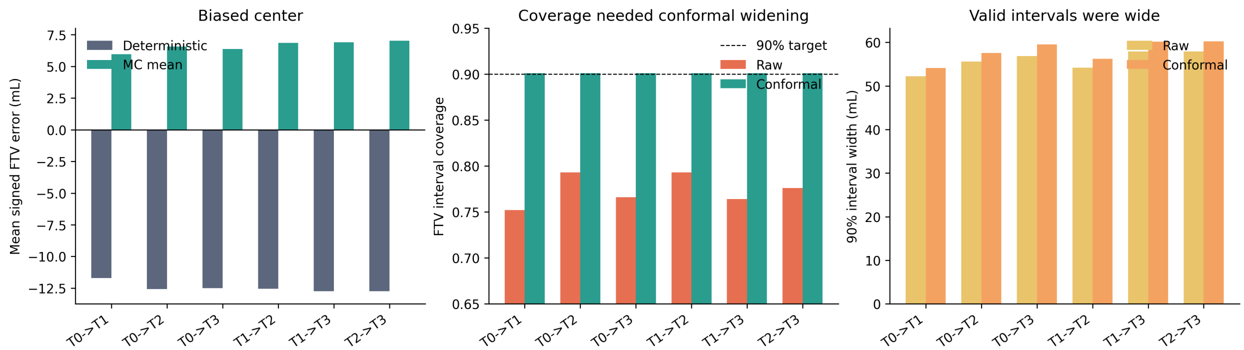

Figure 1: Conditional Monte Carlo diagnostics for the original scheduled-sampling rollout model. The deterministic rollout was biased low for FTV, while the residual Monte Carlo mean overcorrected high. Raw 90% FTV intervals under-covered, and split-conformal expansion restored nominal coverage only with wide intervals.

Table 1: Summary of the original conditional Monte Carlo FTV results. Coverage is for 90% FTV intervals.

Bucket	Obs mean	Det mean	MC mean	Det bias	MC bias	Raw / Conf cov.
T0→T1	24.31	12.61	30.26	-11.70	5.95	0.752 / 0.901
T0→T2	24.75	12.16	31.30	-12.58	6.56	0.793 / 0.901
T0→T3	24.94	12.43	31.30	-12.51	6.36	0.766 / 0.901
T1→T2	24.75	12.18	31.61	-12.56	6.86	0.793 / 0.901
T1→T3	24.94	12.20	31.85	-12.74	6.90	0.764 / 0.901
T2→T3	24.94	12.19	31.97	-12.75	7.02	0.776 / 0.901

The second baseline MC result was that later conditioning helped volume scoring, but did not fully solve spatial uncertainty. T3 FTV CRPS improved as more observed visits were included, from 8.35 for T0-to-T3 to 7.49 for T0,T1,T2-to-T3. However, the corresponding SWD worsened from 1.32 mm to 2.97 mm in the Monte Carlo summaries. This suggested that the sampler was capturing some volume behavior but was still weak as a spatial uncertainty model.

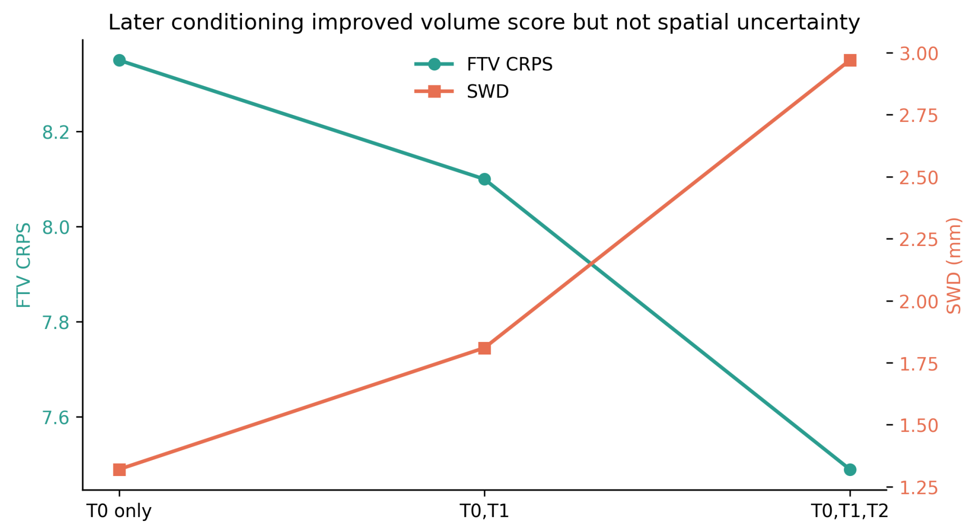

Figure 2: T3 conditional Monte Carlo behavior. Later conditioning improved FTV CRPS, but SWD worsened, showing that volume uncertainty and spatial uncertainty were not improving together.

The third baseline MC result was about the near-zero T3 MRI FTV diagnostic. The field was computed as

$$P(\mathrm{FTV}_{T3} < 0.1\ \mathrm{mL}),$$

which only asks whether the simulated T3 MRI FTV is essentially zero under a fixed threshold. In this run the value was almost always near zero, around 0.007 to 0.009 at T3. This means the threshold is very strict and should be treated as a descriptive imaging diagnostic, not as a central response endpoint. The threshold was not part of the retraining objective; the retrained models learned FTV and alive-burden targets.

The fourth baseline MC result was alive-count miscalibration. The original conditional MC experiment had T3 alive-count absolute errors around 39 to 42 supervoxels. Since alive mass controls which nodes contribute to active tumor volume, this was not a cosmetic issue; it directly affected FTV calibration.

5 Why further exploration was justified

The Monte Carlo experiment changed the interpretation of the earlier success. The model was good enough to demonstrate coherent graph rollout behavior, and the conformal wrapper could recover nominal interval coverage. However, the center of the uncertainty distribution was biologically miscalibrated.

The main lesson was:

good spatial rollout metrics do not guarantee calibrated tumor burden.

The model had learned local graph dynamics, but the clinically important endpoint was not directly controlled. If the deterministic rollout is wrong by about 12 to 13 mL at T3, downstream Monte Carlo response summaries inherit that bias. This motivated a targeted follow-up question:

> Can I keep the same consistent graph rollout structure, but add training pressure so the rollout is also calibrated for FTV and alive tumor burden?

6 Biological retraining response

The retraining grid kept the same consistent graph rollout model family, but added biological endpoint supervision while retaining the scheduled-sampling rollout setup:

- aggregate FTV loss, penalizing patient-level error in total active tumor burden;
- alive supervision, encouraging correct alive/dead supervoxel behavior and alive mass; and
- the same held-out fold protocol used for the original scheduled-sampling model.

This distinction matters. The MC results reported below evaluate the bio retraining checkpoints. Those checkpoints added FTV and alive-burden objectives, but they were not the later random-start-plus-horizon rerun. Therefore, this analysis isolates whether biological endpoint calibration alone fixes the MC calibration problem before changing the rollout-start protocol.

The evaluated variants were:

`bio_ftv000_alive002`	alive-only biological retraining,
`bio_ftv005_alive001`	low FTV + low alive,
`bio_ftv010_alive002`	medium FTV + medium alive,
`bio_ftv020_alive005`	high FTV + high alive,
`bio_ftv010_alive000`	FTV-only ablation.

The retraining results were much better aligned with the Monte Carlo diagnosis. The baseline T0-to-T3 FTV MAE was 12.88 mL. The best retrained candidate, `bio_ftv020_alive005`, reduced this to 5.88 mL, a 54.3% reduction. Its mean T3 FTV prediction was 23.21 mL against an observed mean of 24.94 mL, leaving a mean signed error of -1.73 mL rather than -12.51 mL.

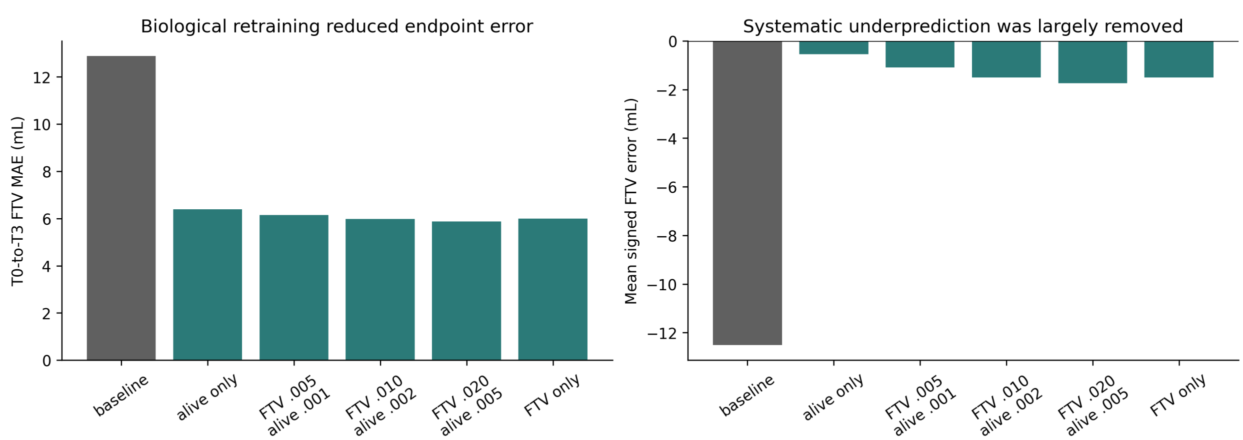

Figure 3: FTV endpoint calibration after biological retraining. The retrained candidates reduced T0-to-T3 FTV MAE by roughly half and removed most of the systematic underprediction seen in the baseline rollout model.

Table 2: T0-to-T3 evaluation of the baseline and biological retraining candidates. Lower FTV MAE is better.

Model	Pred FTV	Obs FTV	Bias	FTV MAE	MAE reduction
`v2_sched_samp`	12.43	24.94	-12.51	12.88	0.0%
`bio_ftv000_alive002`	24.40	24.94	-0.54	6.39	50.4%
`bio_ftv005_alive001`	23.85	24.94	-1.10	6.15	52.2%
`bio_ftv010_alive002`	23.44	24.94	-1.50	5.99	53.5%
`bio_ftv020_alive005`	23.21	24.94	-1.73	5.88	54.3%
`bio_ftv010_alive000`	23.44	24.94	-1.51	5.99	53.5%

The paired patient-level comparison gave the same conclusion. For `bio_ftv020_alive005`, the mean paired FTV MAE improvement was 7.00 mL, with an approximate 95% interval of 6.06 to 7.93 mL, and 79.6% of patients improved.

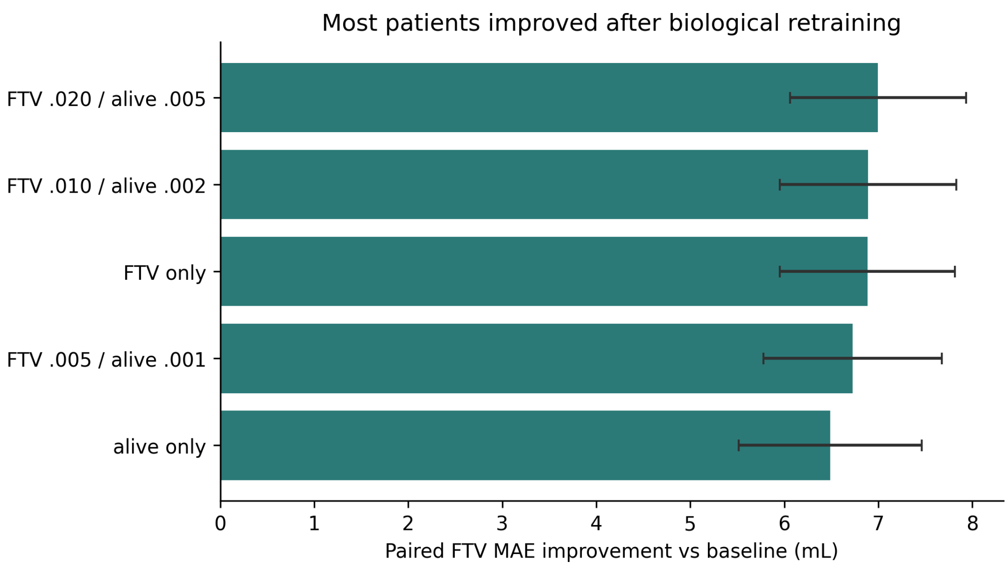

Figure 4: Patient-level paired FTV MAE improvement relative to the baseline scheduled-sampling rollout model. Positive values mean the retrained model reduced FTV absolute error for the same patient.

7 Geometry was preserved

The important caveat is that the retrained models did not simply become crude volume predictors. The spatial metrics stayed essentially flat. For the best FTV candidate, mean SWD changed by about 0.003 mm and Chamfer changed by about 0.006 mm relative to the baseline. Dice changed by less than 0.001.

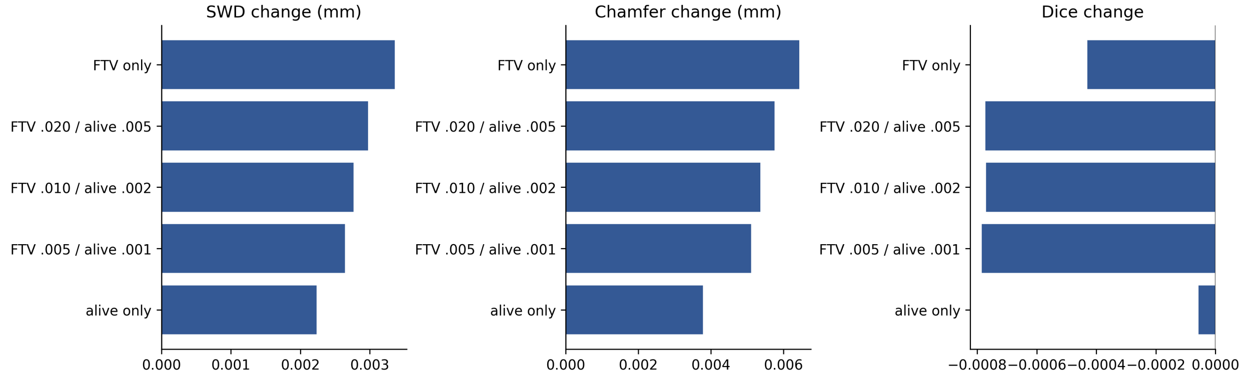

Figure 5: Changes in spatial metrics relative to the original scheduled-sampling baseline. The FTV improvement was achieved without a meaningful collapse in spatial rollout quality.

Alive-count error also improved dramatically in the retraining evaluations, from 58.00 in the baseline simulation evaluation to about 1.70 in the biological retraining variants.

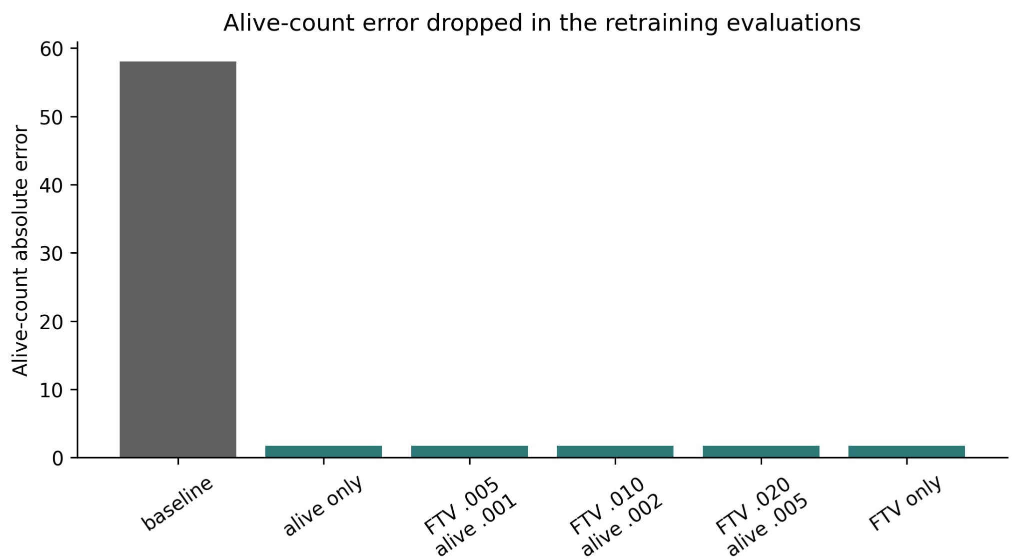

Figure 6: Alive-count absolute error in the baseline and retraining evaluations. The improvement is large, but the current ablation grid does not isolate the explicit alive-loss term as the sole driver because the FTV-only ablation also improved.

The alive term should be interpreted cautiously. The alive-count improvement was identical across the biological retraining variants, including the FTV-only ablation. The current evidence strongly supports endpoint-aware retraining, especially the aggregate FTV objective, but it does not cleanly prove that explicit alive supervision is the mechanism behind the alive-count improvement.

8 Updated conditional Monte Carlo results on retrained models

After the deterministic retraining analysis, I reran the conditional Monte Carlo sampler on three FTV-calibrated checkpoints: `bio_ftv020_alive005`, `bio_ftv010_alive000`, and `bio_ftv010_alive002`. I kept the sampler unchanged: same residual buckets by start and predicted visit, same target-patient exclusion from the calibration pool, same Gumbel top-k alive-mask sampling, same split-conformal interval calculation, same 758-patient cohort, and 256 MC samples per rollout record. This was important because the question was not whether a new sampler could be tuned to look better. The question was whether fixing the deterministic FTV center would make the existing uncertainty distribution better calibrated.

The answer was yes. For the primary model, `bio_ftv020_alive005`, T0-to-T3 deterministic bias stayed close to zero at -1.73 mL instead of returning to the old -12.51 mL bias. The MC mean no longer had to compensate for a badly undercentered base model: T0-to-T3 MC-mean FTV MAE dropped from 15.26 mL to 7.61 mL. Raw 90% FTV coverage improved from 0.766 to 0.876 before conformal correction, and the conformal 90% interval width shrank from 59.51 mL to 27.95 mL. CRPS improved from 8.35 to 5.16, and MC alive-count error fell from 40.78 to 2.69 supervoxels.

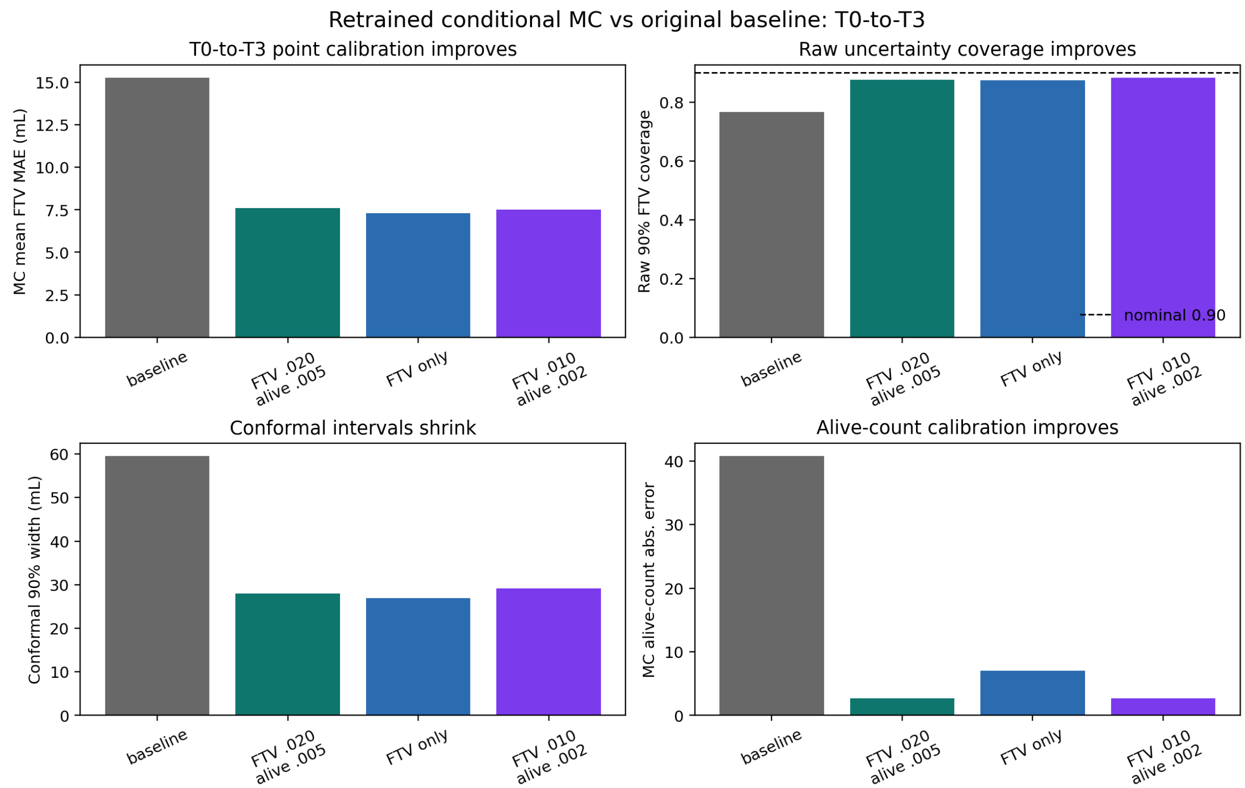

Figure 7: T0-to-T3 conditional MC comparison between the original scheduled-sampling baseline and the three FTV-calibrated retrained models. The retrained models reduce MC-mean FTV error, improve raw 90% coverage, shrink conformal intervals, and greatly reduce alive-count error.

Table 3: T0-to-T3 conditional MC comparison. Coverage is for raw 90% FTV intervals before conformal expansion. Width is the conformal 90% FTV interval width.

Model	Det bias	Det MAE	MC bias	MC MAE	Raw cov.	Width	Alive err.
`v2_sched_samp`	-12.51	12.88	6.36	15.26	0.766	59.51	40.78
`bio_ftv020_alive005`	-1.73	5.88	1.93	7.61	0.876	27.95	2.69
`bio_ftv010_alive000`	-1.51	5.99	1.14	7.31	0.875	26.92	7.00
`bio_ftv010_alive002`	-1.50	5.99	1.76	7.53	0.883	29.14	2.70

The improvement also held across the three T3 conditioning buckets. For `bio_ftv020_alive005`, MC-mean FTV MAE was 7.61 mL for T0-to-T3, 7.18 mL for T1-to-T3, and 5.80 mL for T2-to-T3. The corresponding conformal widths were 27.95 mL, 29.61 mL, and 24.19 mL, all far below the old baseline range of about 59 to 60 mL. This matters because the MC improvement was not only a single T0-to-T3 artifact; it persisted when later observed imaging visits were used as conditioning information.

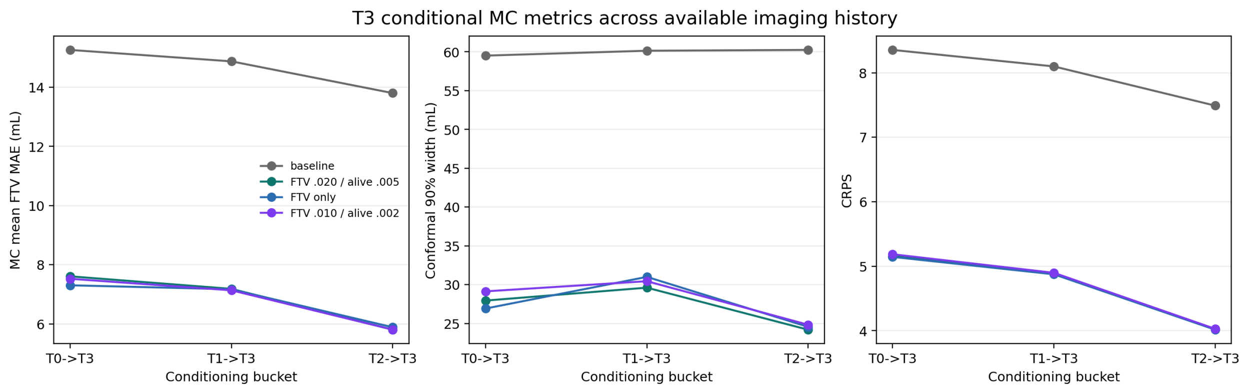

Figure 8: T3 conditional MC metrics across available imaging history. The FTV-calibrated retrained models reduce MC-mean FTV MAE, conformal interval width, and CRPS across T0-to-T3, T1-to-T3, and T2-to-T3 conditioning.

8.1 Patient-level uncertainty visualization

The aggregate metrics are necessary, but they are not enough by themselves. The new patient-level plots show what the MC distribution is doing for individual rollouts. In these plots, the black curve is the observed FTV trajectory, the gray curve and band show the original baseline MC mean plus or minus one standard deviation, and the green curve and band show the retrained primary model. These are MRI FTV uncertainty trajectories from the T0-conditioned rollout.

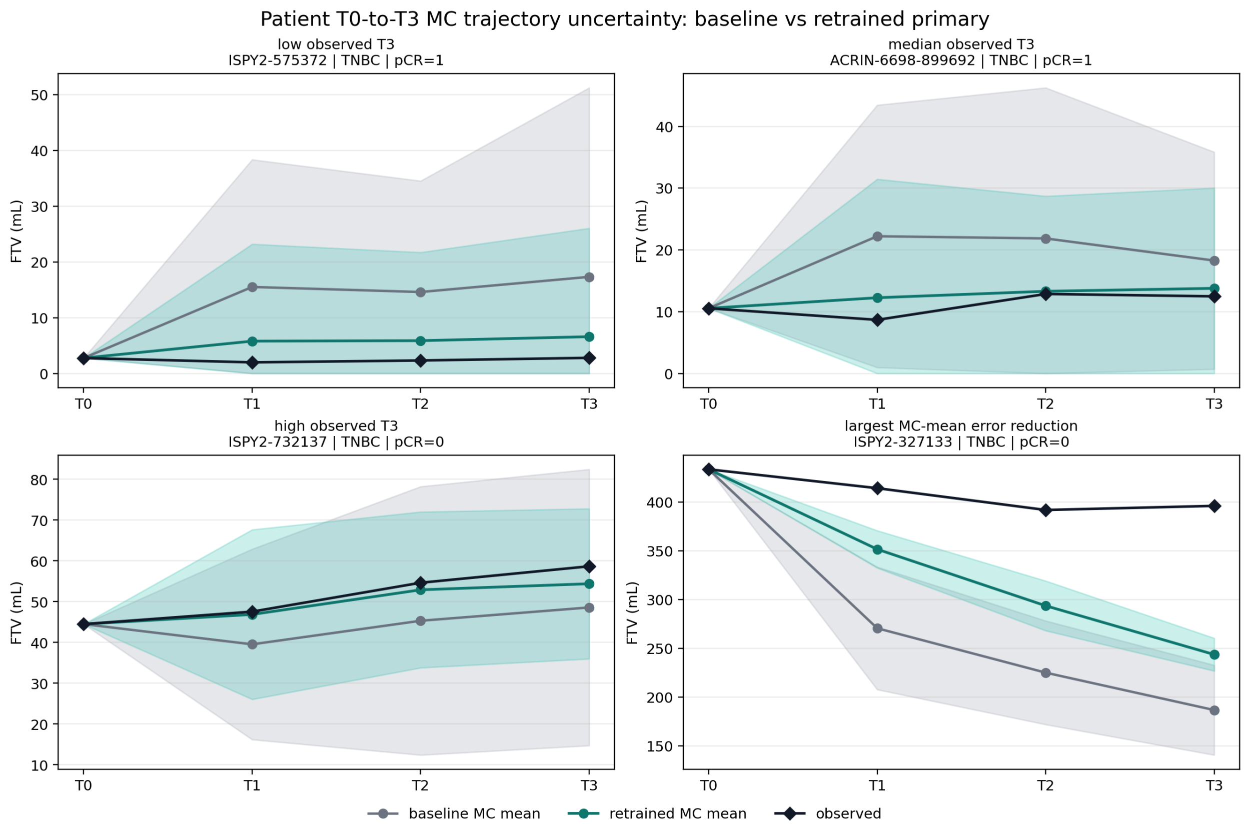

Figure 9: Example patient-level T0-to-T3 MC trajectories. The retrained primary model generally shifts the MC mean closer to the observed FTV path and reduces the width of the one-standard-deviation band. The examples were selected to show low, median, high, and strongly improved T3 cases, so the figure is illustrative rather than a replacement for cohort-level calibration metrics.

The cohort-wide sorted plot gives the same comparison for all 758 patients in the T0-to-T3 setting. Patients are ordered by observed T3 FTV, so this figure shows whether the MC distribution follows the empirical endpoint distribution across low-, moderate-, and high-volume cases. The baseline MC distribution has a visibly wide band and often sits high after residual correction. The retrained model is more centered and sharper. That interpretation is supported by the quantitative results: raw coverage improved from 0.766 to 0.876 while the conformal width shrank from 59.51 mL to 27.95 mL. In other words, the narrower band is not just overconfidence; it is paired with better empirical coverage and lower CRPS.

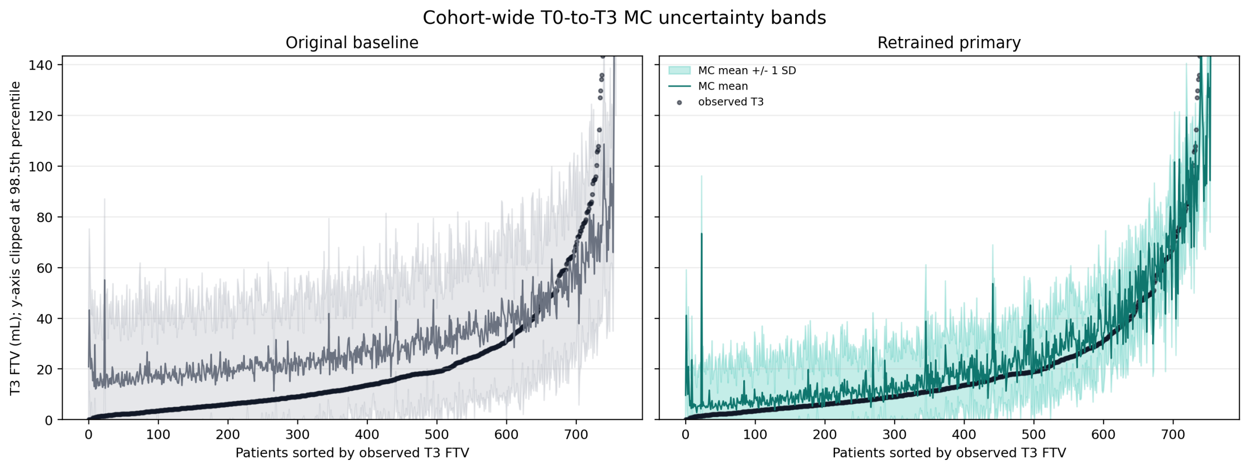

Figure 10: Cohort-wide T0-to-T3 MC uncertainty bands, with patients sorted by observed T3 FTV. The retrained primary model has a narrower and better centered MC distribution across the cohort, while the baseline distribution remains broad and shifted upward after residual correction.

The standard-deviation summary isolates uncertainty width directly. From T0 conditioning, the baseline mean MC FTV standard deviation increased from 24.71 mL at T1 to 27.56 mL at T3. The retrained primary model had substantially smaller standard deviations: 12.60 mL at T1, 15.33 mL at T2, and 16.10 mL at T3. This is exactly the expected pattern if the old MC layer was compensating for a biased deterministic center. Once the deterministic center was corrected, the residual distribution no longer had to spread as widely to cover the observed endpoint.

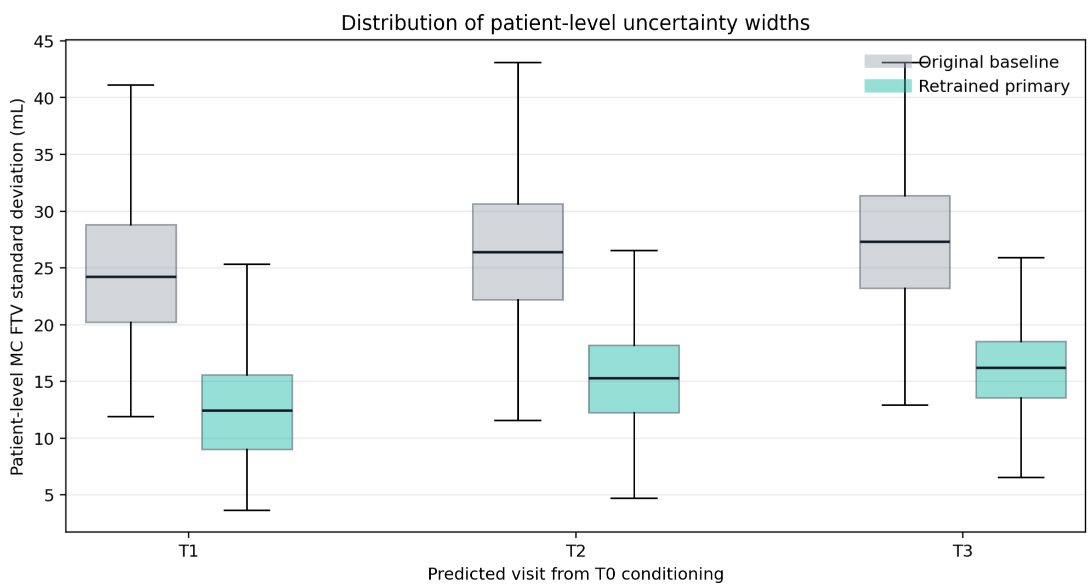

Figure 11: Patient-level MC FTV standard deviation by predicted visit from T0 conditioning. The retrained primary model has consistently lower patient-level uncertainty width than the original baseline, especially at the final T3 endpoint.

All three retrained candidates passed the prespecified T0-to-T3 decision gates. The FTV-only ablation had the lowest T0-to-T3 MC-mean FTV MAE, 7.31 mL, and the narrowest conformal interval width, 26.92 mL. However, its MC alive-count error was 7.00 supervoxels, compared with 2.69 for `bio_ftv020_alive005` and 2.70 for `bio_ftv010_alive002`. I therefore interpret the FTV-only ablation as evidence that aggregate FTV supervision explains much of the volume calibration gain, but I would still keep `bio_ftv020_alive005` as the primary balanced candidate because it combines strong FTV calibration with much better alive-count calibration.

The main residual caution is subgroup drift. For the primary model at T0-to-T3, the HR-/HER2+ subgroup remained the hardest subgroup: deterministic FTV MAE was 8.65 mL, MC-mean FTV MAE was 10.78 mL, raw coverage was 0.842, and CRPS was 8.07. The near-zero T3 MRI FTV diagnostic remains an imaging-threshold summary, not a primary endpoint in this analysis.

9 Progress interpretation

The concise interpretation is:

> I had a model that worked well as a rollout model, but when I used it as the center of a conditional Monte Carlo simulator, I found a systematic biological calibration problem. It underpredicted T3 FTV by about 12.5 mL. The residual MC layer could recover nominal coverage after conformal expansion, but the intervals were wide and the MC mean overcorrected. That told me the uncertainty method was compensating for a biased base model.
>
> Because FTV is the key imaging endpoint for tumor burden, I retrained the same consistent graph model with aggregate FTV and alive-burden losses. The best retrained model cut T0-to-T3 FTV MAE from 12.88 mL to 5.88 mL while preserving the spatial metrics. When I reran the same conditional MC sampler on the retrained models, the uncertainty distribution also improved: the primary model cut T0-to-T3 MC-mean FTV MAE from 15.26 mL to 7.61 mL, increased raw 90% coverage from 0.766 to 0.876, and reduced conformal interval width from 59.51 mL to 27.95 mL. The follow-up was therefore not a random model tweak; it was a targeted response to a failure mode revealed by the Monte Carlo simulation, and the completed MC rerun shows that correcting the deterministic FTV center materially improved uncertainty calibration.

10 Current conclusion and next step

The current conclusion is that `bio_ftv020_alive005` is the best balanced candidate from this phase. The FTV-only ablation slightly wins on T0-to-T3 MC-mean FTV MAE and conformal width, but the high-FTV/high-alive model has much better alive-count calibration while still passing every FTV and uncertainty gate. That balance matters because the model is supposed to remain a graph rollout model, not only a scalar FTV predictor. The patient-level trajectory plots make the same point visually: the retrained model produces a more credible FTV uncertainty distribution because it is simultaneously closer to the observed trajectory, narrower, better covered, and lower by CRPS.

The next methodological decision is whether the corrected random-start-plus- horizon model family should replace this scheduled-sampling bio grid as the final architecture. If so, the same MC experiment should be repeated on that checkpoint family, using the same cohort and sampler, so that the comparison is not confounded by changes in the uncertainty machinery. Separately, the subgroup results should be kept visible: HR-/HER2+ patients remain the hardest group in the primary T0-to-T3 MC run, so that subgroup needs targeted error review before making strong claims about uniform calibration.

The current MC output supports calibrated T3 MRI FTV uncertainty. Any separate clinical outcome model should be treated as a distinct prediction problem built from appropriate outcome labels and covariates, rather than folded into this rollout uncertainty analysis.